

Notes on Incremental Typesetting

Jordan Example

September 2026

1 Introduction

A live preview must answer a keystroke faster than the reader can notice. The usual approach — recompile the whole document on every change — is correct but slow, and it also loses the reader’s place on the page. These notes describe the two mechanisms that avoid both problems: a block cache keyed by source bytes, and a page builder that replays TeX’s cost model.

We assume familiarity with the standard L^AT_EX document model: a preamble, a sequence of sectioning commands, paragraphs, display material and floats. Nothing here depends on a particular class; the examples use `article` because it is the class of most of the corpus.

2 Method

2.1 Block cache

Each paragraph is hashed together with the font state at its first character. A cache hit returns the shaped glyph runs and the chosen line breaks; a miss reshapes and re-breaks only that paragraph. The cost of a miss is therefore bounded by the longest paragraph, not by the document.

2.2 Page builder

Blocks are contributed to the current page in order. With the natural height h_i and following penalty p_i of block i , the badness of breaking after block n on a page of goal height G is

$$b_n = \min \left(10000, 100 \left(\frac{G - \sum_{i=1}^n h_i}{s} \right)^3 \right), \quad (1)$$

where s is the available stretch. The break with the least total cost $b_n + p_n$ is taken, exactly as in the batch compiler, so a cached block lands on the same page it would have in a fresh run.

2.3 What is not cached

Floats, footnotes and anything that changes a counter are recomputed on every keystroke. This is deliberate: their placement depends on the pages around them, and a stale placement is worse than a slow one.

3 Related work

The line-breaking algorithm follows Knuth and Plass [2]; the page builder follows the description in *The T_EXbook* [1]. Incremental recomputation over a block cache is standard in build systems and editors [3]; what differs here is that the cost model being replayed is TeX’s own.

4 Conclusion

The two mechanisms introduced in Sections 1 and 2 are independent: the cache can be disabled without changing output, and the page builder can be driven by a batch compiler. That separation is what makes the pixel comparison in the appendix meaningful.

A Measurement protocol

Every document is compiled twice by pdfLaTeX with a fixed `SOURCE_DATE_EPOCH`, then rasterised at 144 dpi. The producer’s PDF is rasterised the same way and differing pixels are counted per page. Page counts are compared first; a page-count mismatch is reported as such and no pixel count is claimed for the missing pages.

References

- [1] Donald E. Knuth. *The T_EXbook*. Addison-Wesley, 1984.
- [2] Donald E. Knuth and Michael F. Plass. Breaking paragraphs into lines. *Software: Practice and Experience*, 11(11):1119–1184, 1981.
- [3] Matthew A. Hammer et al. Adapton: composable, demand-driven incremental computation. In *PLDI*, 2014.